

Re: Manuscript for your Review – Vessel Diameter Underestimation in OCTA



From Rene Werkmeister <rene.werkmeister@meduniwien.ac.at>
To Yash Patel <yash.patel@meduniwien.ac.at>
Cc Conrad Merkle <conrad.merkle@meduniwien.ac.at>
Date 04-09-2026 23:22

Dear Yash,

thank you again for this very interesting manuscript and for including me in this research.

Below a few comments...and sorry for really making full use of the deadline until the very last hours of September,4th :)

I m not a big fan of the comment functions in Adobe, I find it gets confusing quite quickly, so I have written my comments directly below with reference to the respective line numbers. I hope they are helpful:

Line 6: if possible, could you please include the initial of my middle name, thus "René M. Werkmeister"

Line 21: "Before contrast"...IMHO "Before contrast enhancement" would be more clear

In general: Was the degree of underestimation somehow related to vessel diameter? Since OCTA detects mainly the RBC column, while the Intralipid "ground truth" also includes the cell-free layer close to the vessel wall, I would expect the underestimation to be more pronounced for smaller vessels, where the cell-free layer accounts for a larger fraction of the total vessel diameter. Could be interesting to check/discuss this.

Line 27 (minor comment): US-english would be "characterize"

Lines 30-36: Is this intended as a general statement or specifically in the context of OCTA? I think the dependence of vessel density on vessel caliber applies when vessel density is defined as the "vessel-occupied area" in the angiogram. For skeleton/length-based density metrics, vessel diameter should not have the same direct effect. Maybe worth specifying this here?

Line 32 (minor): "These findings" sounds IMHO a bit odd here...maybe "These vascular changes are commonly quantified..."?

Line 60 (minor): "binarization" in US english...Seems you are using British english – shouldnt we switch to US English for BiomedOptEx?...if intended, please ignore my comment :)

Line 71: "underestimation"...(deleted (s))

Line 109 (minor): Maybe "during Intralipid injection" or "contrast agent injection" instead of "contrast injection"?

Line 119: Since this was performed in only one mouse, should we frame this more explicitly as an exploratory "proof-of-concept experiment"? "To assess whether..." sounds a bit strong for n=1...One could write sth like: "As an exploratory proof-of-concept, functional flicker stimulation was performed in one C57BL/6J mouse to examine how the choice of diameter measurement method influences the observed vascular response."

Line 125: Is it really the OCT signal that needs to stabilize here, or rather the vascular response/vasodilation following stimulus onset?... "OCT signal to stabilize" IMHO sounds a bit like a technical limiation

Line 161 (minor): Maybe "following the arrival of the Intralipid bolus" instead of "after arrival of contrast"?

Line 181: Does this really ensure that the measured diameter reflects the "true" vessel size, or does it specifically correct for the geometric effect of off-axis cross-sectioning? ...of course I know what you mean here but it sounds quite strong. I would maybe phrase like: "...to correct the measured vessel diameters for the effect of off-axis cross-sectioning."

In general: although, the Gaussian fitting has been used in previous studies for vessel diameter measurements, should we explain why a Gaussian/GG is an appropriate model for the OCTA vessel intensity profile and why its width can be interpreted as vessel diameter?...from a "logical standpoint", one would expect the GG model to work better, since for well-resolved vessels (especially larger vessels), one would expect a more flat-topped profile (as can also be seen in the data). Maybe this is discussed later in the text anyway – just wanted to leave a comment here so I don't forget :)...ok, found it in 378-386 in discussion :)

Line 229: Is the regression constrained to pass through zero? Otherwise, wouldnt perfect agreement require both m = 1 and an intercept of 0?...I would clarity this (ok, in the graphs I see what you did :))

Line 237: I am a bit worried about the paired t-test here since the flicker experiment was performed in only 1 mouse. If the individual vessels are treated as the paired observations, can we really use this to infer statistical significance, given that all vessels come from the same animal/circulation? Maybe this is fine for the proof-of-concept, but I just wanted to check...to be honest, I would just report these changes descriptively without t-test (but since it is not the major point of the paper, I don't think a reviewer would consider this a major issue.)

Line 263ff: I find this section quite difficult to follow with all the different combinations of projection methods, diameter metrics, contrast conditions, and outcome measures. ...but I understand you have many different approaches and it is difficult to summarize...and it is much easier for me to make this comment than to implement it. Thus, sorry for the following: Maybe, if possible, one could shorten it a bit and focus more on the main "take-home messages" and leave some of the individual numbers/comparisons to the table and figures? I think there is also some repetition (e.g. the underestimation with wFWHM,GG and the overall MIP vs MeLP comparison are mentioned at several points) but this is a minor point cause it bringsup the main findings again...butin general at the moment I find it a bit difficult to keep track of the main message.

Line 310ff: As mentioned above, I am still a bit unsure about the significanc testing given that this is only 1 mouse. Also, if the different time points within each vessel are treated as independent observations, couldnt this result in the very small pvalues (e.g. p < 0.001) even though they are not really independent? I think the results are quite convincing even if reported descriptively without the t-test.

Line 325: Could the gradual decrease in angiogram intensiyt over the scan be related to a decrease in Intralipid contrast enhancement after the bolus? If so, could this also contribute to the apparent vasoconstriction seen with the binary-mask-based method?...again, only a comment also for myself...maybe it anyway comes in the discussion

Line 342: Not sure if I would call both methods "threshold-dependent" here in this context. The model-based methods use a defined relative width (e.g. $1/e^2$ or FWHM), but they don't depend on an absolute intensity threshold in the same way as the BM method, right?

Line 348: would not call it "systematic offset" as this would be an additive intercept. Maybe better "systematic proportional bias" (?)

Line 352: IMHO "makes them usable even pre-contrast" sounds a bit strong? The relationship is more linear/predictable, but there is still substantial diameter underestimation pre-contrast. Maybe phrasing a bit more cautiously, like "more predictable/systematic even pre-contrast" (?)

Line 413: Do we know as a fact that the outlier sensitivity actually becomes the "dominant source of error" here? The explanation makes sense of course, but maybe this is a bit too strong as we have not directly tested the mechanism. Maybe something like "may contribute more strongly to the measurement error" (?)

Line 436 (minor): "MeIP should be the preferred projection method" sounds quite general/strong. Maybe a bit weaker, e.g. "for quantitative diameter measurements using the methods evaluated here"

Line 445: As mentioned above, could the decrease in angiogram intensity also be related to the Intralipid contrast gradually decreasing after the bolus?

Line 447: IMHO, these statements might be a bit too strong. Instead of "biases" and "remains unaffected" maybe "was associated with changes in the binary-mask-based measurements, while the model-based measurements were less affected" (?)

Line 466: Maybe I am overthinking this statement, but SD/CoV mainly supports the "most repeatable" part, right? The agreement with ground truth you show elsewhere, so maybe just separate the two here: "... indicating that it is the most repeatable of the metrics tested. It also showed the closest agreement with ground truth." (?)

Line 510: Not sure if relative comparisons between timepoints are always valid without correction (?) Especially since our flicker experiment shows that changes in signal intensity can affect the diameter measurement. Maybe "may remain valid" would be a bit safer?

Line 529ff (minor): Maybe this section is a bit long, especially given that the flicker experiment is more of a proof-of-concept? Could one shorten the literature comparison a little?

Line 571: Not sure I understand this hypothesis: why would the model-based fit be specifically more sensitive to smooth-muscle-driven dilation in arteries? Is there any evidence/reason to expect this?

All the best,

René

Am 26.08.26 um 17:44 schrieb Yash Patel:

Dear all,

Please find attached the manuscript, "Characterising measurements of vessel diameter in optical coherence tomography angiography and correcting artefact-induced underestimation."

I would be grateful if you could review the manuscript and send me any edits, comments, or suggestions by Friday, 4 September, so that I can incorporate the feedback and prepare the final version.

Thank you all for your contributions and for taking the time to review the manuscript.

Best regards,
Yash

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